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Bichromatic Boundary-State Reduction for Acyclic 4-Edge-Coloring of Outerplanar Graphs with Maximum Degree at Most Four

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Abstract

Determining whether an outerplanar graph of maximum degree four admits an acyclic four-edge coloring requires controlling bichromatic cycles created during graph composition. We develop an exact, constructive dynamic-programming framework based on exposed bichromatic boundary signatures, which record only terminal-to-terminal two-color connections that remain relevant to subsequent series and parallel operations. We prove exact transition rules, establish contextual equivalence through separating path contexts, and characterize one-sided contextual dominance by set inclusion. These results justify lossless removal of dominated feasible signatures while preserving a reconstructible coloring. For a fixed four-color palette, the formal state space contracts from 16,384 full signatures to 956 exposed signatures; inclusion-antichain bounds limit undominated families to 468 states, and color symmetry yields 82 formal orbits. Given linear-size series–parallel decompositions, the algorithm decides feasibility and reconstructs an acyclic coloring in linear time. We additionally certify an edge-minimal seven-vertex obstruction and derive sharp four- versus five-color thresholds for maximal outerplanar graphs, together with a blockwise criterion for block-maximal graphs. Experiments on 649 benchmark entries show agreement among three algorithmic implementations and no state-consistency or reconstruction errors. At 192 vertices, dominance reduction decreases retained states by 13.68% and peak traced memory by 13.21%, but increases execution time in the evaluated implementation. These structural characterizations do not extend to arbitrary degree-four outerplanar graphs.

KEYWORDS: Acyclic edge coloring; outerplanar graphs; exact dynamic programming; bichromatic connectivity; contextual dominance

1. Introduction

Acyclic edge coloring assigns distinct colors to adjacent edges and forbids cycles using only two colors [1,2]. For outerplanar graphs, the general upper bound is one color above the maximum degree [3–8], but deciding when four colors suffice at maximum degree four requires more detailed structure. The difficulty is compositional: two individually valid colorings can create a forbidden bichromatic cycle when joined. This motivates an exact boundary representation rather than a greedy local coloring rule.

Previous work covers acyclic edge-coloring bounds [9 – 11], related outerplanar coloring notions [12 – 18], and recursive or logic-based graph algorithms [19 – 23]. In particular, Hou and Wu [8] corrected a claimed characterization in [7] by exhibiting infinitely many outerplanar graphs with $\Delta(G) = 4$ and $a'(G) = 5$ that contain no subgraph isomorphic to Q. Their erratum explicitly noted that determining the acyclic edge chromatic number of outerplanar graphs with maximum degree four remained open. We address this problem algorithmically rather than through a forbidden-subgraph characterization. Theorem 17 provides an exact recognition and reconstruction algorithm for acyclic four-edge colorability of outerplanar graphs with maximum degree at most four. In particular, when $\Delta(G) = 4$, the bounds $4 \leq a'(G) \leq 5$ imply that this decision determines $a'(G)$ exactly. Proposition 18 establishes linear time and space when linear-size valid two-terminal series–parallel decompositions are supplied. Our contribution is not a new generic linear-time bounded-treewidth algorithm; it is a problem-specific, provably lossless representation of bichromatic connections and a dominance rule that permits exact constructive state reduction.

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Figure 1 summarizes the distinction between full and exposed boundary information. We focus on exact feasibility, reconstructible witnesses, and checkable structural certificates rather than estimating the number of distinct colorings.

The principal contributions are: (i) exact series/parallel transition rules for exposed signatures and a contextual separation theorem; (ii) a necessary-and-sufficient inclusion criterion for contextual dominance, yielding the formal bounds 16,384 → 956 and ≤468 undominated states; (iii) a linear-time exact recognition-and-reconstruction algorithm for fixed four colors, with reproducible cross-checks against exhaustive search; and (iv) a certified seven-vertex obstruction, a sharp threshold for maximal outerplanar graphs, and an exact blockwise criterion. Each claim is scoped to the graph classes and context models specified below.

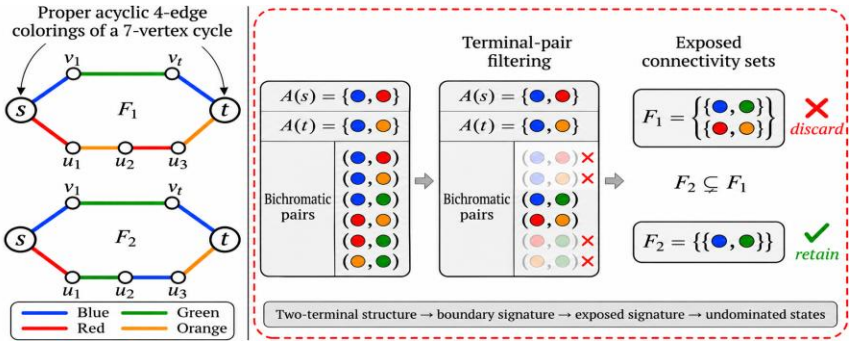


Figure 1. Reduction principle: terminal color sets and relevant two-color paths determine an exposed signature; dominated feasible signatures can be removed without changing four-color feasibility.

2. Basic Graph Structures and Boundary Signatures

2.1. Basic Definitions and Block Decomposition

Throughout this paper, all graphs are finite, simple, and undirected. Let $G = (V, E)$ be a graph with maximum degree $\Delta(G)$. A proper edge coloring is a mapping $c: E \to \{1,2,\dots,k\}$ such that any two adjacent edges receive distinct colors. The coloring c is called *acyclic* if every cycle of G contains at least three distinct colors. The minimum number of colors required for an acyclic edge coloring of G is denoted by $a'(G)$.

For two distinct colors α and β, let $G[\alpha, \beta]$ denote the spanning subgraph of G consisting of all edges assigned either α or β. Since c is proper, every vertex of $G[\alpha, \beta]$ has degree at most two. Consequently, c is acyclic if and only if $G[\alpha, \beta]$ is a forest for every pair of distinct colors.

In the following analysis, the color palette is fixed as $K = \{1,2,3,4\}$. A graph is said to be *acyclically 4-edge-colorable* if it admits an acyclic edge coloring using colors from K. The coloring is not required to use all four colors.

A *block* of G is a maximal subgraph without a cut vertex, where each bridge is treated as a block consisting of a single edge. Isolated vertices are disregarded in the block decomposition. The block decomposition of a graph into nontrivial blocks and bridge blocks is illustrated in Fig. 2.

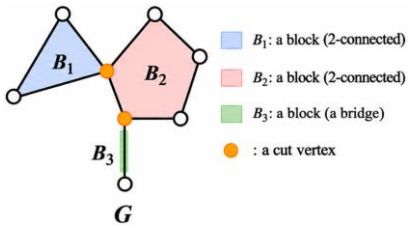


Figure 2. Block decomposition of a graph G, where B_1 and B_2 are 2-connected blocks, B_3 is a bridge block, and orange vertices indicate cut vertices.

Lemma 1 (Block Reduction). *Let G be a graph and let $k \geq \Delta(G)$. Then G admits an acyclic k-edge coloring if and only if every block of G admits an acyclic k-edge coloring.*

Proof. Necessity is immediate. For sufficiency, root the block–cut tree and attach one colored block at a time. At a shared cut vertex, the incident colors in the new block can be permuted to avoid those already used, because the total vertex degree does not exceed the palette size. Every cycle lies entirely inside one block, so the combined coloring remains acyclic. □

2.2. Two-Terminal Boundary Signatures

A *two-terminal graph* is a graph H equipped with two distinct designated vertices s and t, called its *terminals*. Two standard composition operations are considered.

In a **series composition**, two terminal graphs H_1 and H_2 are joined by identifying the second terminal of H_1 with the first terminal of H_2. The remaining terminals become the terminals of the resulting graph.

In a **parallel composition**, the first terminals of H_1 and H_2 are identified, and their second terminals are also identified. The two components are required to be internally vertex-disjoint. Only compositions that produce a simple graph are considered.

The series and parallel composition operations are illustrated in Fig. 3, where the corresponding terminal vertices are identified.

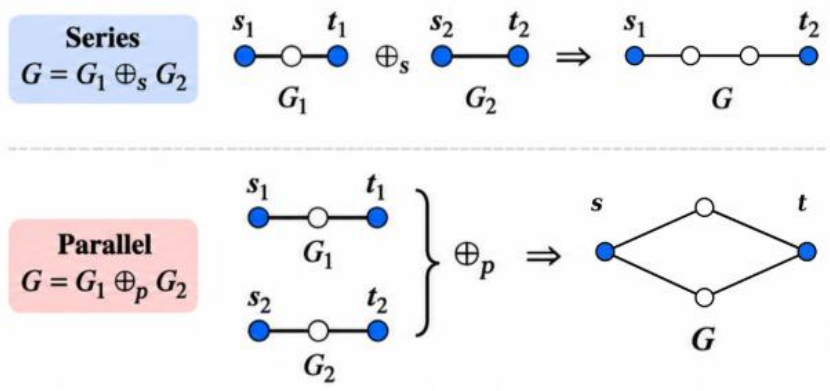


Figure 3. Series composition obtained by identifying t_1 with s_2, and parallel composition obtained by identifying s_1 with s_2 and t_1 with t_2.

A two-terminal series–parallel graph is obtained recursively from single-edge graphs through these operations. Every 2-connected outerplanar graph admits such a decomposition with suitable terminals. Classical algorithms for recognizing related series–parallel structures were developed by Valdes et al. [21]. Together with Lemma 1, this provides the structural basis for the boundary-state analysis.

Let H be a two-terminal graph with terminals s and t, and let c be an acyclic 4-edge coloring of H.

For each terminal $x \in \{s, t\}$, define its *incident color set* by

$$A_c(x) = \{c(e) : e \text{ is incident with } x \text{ in } H\}.$$

For every pair of distinct colors $\alpha, \beta \in K$ with $\alpha < \beta$, define the *bichromatic connectivity indicator*:

$$R_c(\alpha, \beta) = \begin{cases} 1, & \text{if } H[\alpha, \beta] \text{ contains an } s\text{–}t \text{ path,} \\ 0, & \text{otherwise.} \end{cases}$$

Here, an s–t path contained in $H[\alpha, \beta]$ is allowed to use only one of the two colors. This convention is essential for correctly handling paths consisting of a single edge.

Definition 2 (Bichromatic Boundary Signature). *The bichromatic boundary signature associated with an acyclic 4-edge coloring c of H is the tuple*

$$\Sigma(H,c) = (A_c(s), A_c(t), R_c),$$

Definition 3 (Feasible Signature Family). *The feasible signature family of H is defined as*

$$\mathcal{S}_4(H) = \{\Sigma(H,c): c \text{is an acyclic 4-edge coloring of} H\}.$$

For a single edge st colored $\gamma \in K$, the corresponding signature satisfies

$$A_c(s) = A_c(t) = \{\gamma\},$$

and $R_c(\alpha,\beta) = 1$ precisely when $\gamma \in \{\alpha,\beta\}$. Consequently, a single-edge graph has four feasible signatures, one for each choice of γ.

3. Exact Exposed-State Reduction and Contextual Dominance

3.1. Exposed Bichromatic Connectivity

Only bichromatic terminal paths that an admissible external component can complete into a cycle matter for future composition. If both colors of a pair already occur at a terminal, an external path using that pair cannot enter through that terminal without violating properness. This observation yields a smaller signature.

Let $K = \{1,2,3,4\}$ and let $\mathcal{P} = \{\{\alpha,\beta\}: \alpha,\beta \in K,\ \alpha < \beta\}$. For an acyclic edge coloring c of a two-terminal graph H, set $A = A_c(s)$ and $B = A_c(t)$. For a color pair $p \in \mathcal{P}$, define

$$\eta_A(p) = \begin{cases} 1, & |A \cap p| = 1, \\ 0, & \text{otherwise}, \end{cases}$$

and let

$$M(A,B) = \{p \in \mathcal{P}: \eta_A(p) = \eta_B(p) = 1\}.$$

Definition 4 (Exposed Bichromatic Signature). *The exposed connectivity set and the reduced boundary signature of (H,c) are, respectively,*

$$F(H,c) = \{p \in M(A,B): R_c(p) = 1\},$$

$$\rho(H,c) = \big(A,B,F(H,c)\big) \tag{1}$$

The reduced signature discards terminal-blocked pairs but retains exactly those exposed connections needed by the composition tests. Let the feasible reduced-signature family consist of all such signatures realized by acyclic four-edge colorings.

3.2. Exact Composition and Contextual Distinguishability

Theorem 5 (Exact Reduced Composition). *Suppose H_1, H_2 have acyclic 4-edge colorings with reduced signatures $\rho_i = (A_i, B_i, F_i),\ i = 1,2.$*

For their series composition, the union coloring is proper and acyclic precisely when $B_1 \cap A_2 = \varnothing$. The resulting signature is

$$\rho_S = (A_1, B_2, F_1 \cap F_2). \tag{2}$$

For their parallel composition, the union coloring is proper and acyclic precisely when

$$A_1 \cap A_2 = B_1 \cap B_2 = \varnothing, \qquad F_1 \cap F_2 = \varnothing. \tag{3}$$

Writing $A = A_1 \cup A_2$ and $B = B_1 \cup B_2$, the resulting signature is

$$\rho_P = \big(A, B, (F_1 \cup F_2) \cap M(A,B)\big). \tag{4}$$

Proof. Series composition has a single shared cut vertex, so it creates no new cycle; properness requires disjoint incident-color sets there. A two-color terminal path traverses the resulting series graph exactly when both component paths exist, giving Eq. (2).

In parallel composition, a new bichromatic cycle exists exactly when the two components have terminal-to-terminal paths in the same exposed color pair. Distinct incident colors at both common terminals ensure properness; therefore Eq. (3) characterizes feasibility. Any resulting terminal path stays in one component, giving Eq. (4). □

A fixed-colored two-terminal context is a series–parallel expression with one hole and fixed acyclic colorings elsewhere. The context accepts a substitution when the resulting coloring is proper and acyclic, with no recoloring. Comparisons use contexts admissible for both substitutions; they need not preserve outerplanarity, although the resulting graphs must remain simple and series–parallel.

Theorem 6 (Exact Contextual Equivalence and Path Separation). *Let* (H_1, c_1) *and* (H_2, c_2) *be acyclic 4-edge-colored two-terminal series--parallel graphs whose terminal color sets are the same ordered pair* (A, B)*, and let their exposed signatures be* $\rho_i = (A, B, F_i)$*,* $i = 1,2$*. The following statements are equivalent:*

(i) $F_1 = F_2$.

(ii) Every common admissible fixed-colored two-terminal series--parallel context accepts both substitutions or rejects both substitutions.

Moreover, if $F_1 \neq F_2$, there exists a separating common admissible context consisting of a single internally vertex-disjoint path of length two or three attached in parallel. For every $p \in F_1 \triangle F_2$, this path can be colored using exactly the two colors of p so that the resulting context accepts precisely the substitution for which $p \notin F_i$. The separating construction lies within the class of simple two-terminal series--parallel graphs; no assertion is made that both resulting graphs are outerplanar.

Proof. If two exposed signatures agree, Eq. (2)–(4) propagate identical feasibility decisions and resulting signatures at every context node; hence no common admissible fixed-colored context distinguishes them.

Conversely, suppose that $F_1 \neq F_2$ and choose $p = \{\alpha, \beta\} \in F_1 \triangle F_2$. Because $F_i \subseteq M(A,B)$, the pair p meets each of A and B in exactly one color. Let γ_s and γ_t be the unique colors in $p \backslash A$ and $p \backslash B$, respectively, ensuring that the new path edges incident with s and t use colors absent from the corresponding terminal color sets. Introduce new internal vertices and join the terminals s, t by a path P whose internal vertices are disjoint from both substituted graphs. If $\gamma_s \neq \gamma_t$, take a two-edge path with consecutive colors (γ_s, γ_t). If $\gamma_s = \gamma_t$, write $p = \{\gamma_s, \delta\}$ and take a three-edge path with colors $(\gamma_s, \delta, \gamma_s)$. In either case, P is properly colored and its unique terminal-to-terminal path uses exactly the colors in p. Hence its exposed connectivity set is $\{p\}$.

At s and t, the new incident colors are absent from A and B, respectively. Parallel attachment is therefore proper for both substitutions. The added internal vertices guarantee simplicity, and both resulting graphs are two-terminal series--parallel. By Theorem 5, the parallel union is acyclic exactly when $F_i \cap \{p\} = \emptyset$, that is, exactly when $p \notin F_i$. Since p belongs to exactly one of F_1, F_2, the two substitutions are distinguished. Notice also that $p \in M(A,B)$ implies $|A|, |B| \leq 3$; the attachment therefore respects the degree-four bound at both terminals. This proves the converse and the stated path-separation property. □

3.3. Exact Contextual Dominance and Minimal Feasible States

For feasible reduced signatures $\rho_i = (A, B, F_i)$ with the same ordered terminal color sets, define the inclusion relation

$$\rho_1 \preccurlyeq \rho_2 \quad \Leftrightarrow \quad F_1 \subseteq F_2. \tag{5}$$

The significance of this relation is not merely combinatorial. It exactly characterizes the one-sided ability of a fixed-colored series--parallel context to accept one state whenever it accepts another.

Theorem 7 (Exact Characterization of Contextual Dominance). *Let* (H_1, c_1) *and* (H_2, c_2) *be acyclic 4-edge-colored two-terminal series--parallel graphs with reduced signatures* $\rho_1 = (A, B, F_1)$ *and* $\rho_2 = (A, B, F_2)$. *The following statements are equivalent:*

(i) $\rho_1 \preccurlyeq \rho_2$, that is, $F_1 \subseteq F_2$.

(ii) For every common admissible fixed-colored two-terminal series--parallel context $\mathcal{C}$, acceptance of (H_2, c_2) by $\mathcal{C}$ implies acceptance of (H_1, c_1) by $\mathcal{C}$.

Furthermore, if $F_1 \subsetneq F_2$, there is a common admissible two- or three-edge parallel-path context that accepts (H_1, c_1) and rejects (H_2, c_2). Thus the preorder in (5) is exactly the contextual acceptance preorder for signatures with fixed terminal color sets.

Proof. Let the smaller exposed set be contained in the larger one, with identical terminal color sets. Process the fixed context from the hole to its root. Series composition preserves inclusion by intersection, while parallel composition preserves inclusion by union and cannot introduce a forbidden intersection for the smaller set if none exists for the larger set.

The assertion holds at the hole by hypothesis. Suppose it holds immediately before composition with a fixed-colored component having signature $\rho_C = A_C, B_C, F_C$. For a series composition, compatibility depends only on disjointness of the relevant terminal color sets, which are identical for the two substitutions. Therefore both compositions are compatible, and their new exposed sets satisfy

$$F_1 \cap F_C \subseteq F_2 \cap F_C.$$

The same argument applies regardless of whether the fixed component is placed on the left or the right. For a parallel composition, the terminal color-disjointness conditions again coincide. Since the composition containing the second substitution is acyclic, $F_2 \cap F_C = \emptyset$. It follows from $F_1 \subseteq F_2$ that $F_1 \cap F_C = \emptyset$, so the composition containing the first substitution is also acyclic. Writing (A', B') for their common resulting terminal sets, the exposed sets satisfy

$$(F_1 \cup F_C) \cap M(A', B') \subseteq (F_2 \cup F_C) \cap M(A', B').$$

Inductively, whenever the context accepts the larger state, it also accepts the smaller one. This proves sufficiency.

For the converse, suppose $F_1 \not\subseteq F_2$. Choose $p \in F_1 \setminus F_2$. By the separating-path construction in the proof of Theorem 6, there is a common admissible fixed-colored parallel-path context that accepts (H_2, c_2), because $p \notin F_2$, but rejects (H_1, c_1), because $p \in F_1$. This contradicts (ii). Hence (ii) implies $F_1 \subseteq F_2$. Finally, when $F_1 \subsetneq F_2$, applying the same construction to any $p \in F_2 \setminus F_1$ gives a context that accepts the first substitution and rejects the second. □

Definition 8 (Minimal Feasible Signature Family). *For a two-terminal series--parallel graph* H, *let* $\mathcal{R}_4(H)$ *denote its feasible reduced-signature family. Its undominated family is*

$$\mathcal{R}_4^{\min}(H) = \{(A,B,F) \in \mathcal{R}_4(H) : \nexists\, (A,B,F') \in \mathcal{R}_4(H) \text{with} F' \subsetneq F\}. \quad (6)$$

For fixed terminal sets, feasible exposed sets form a finite inclusion poset; keep only its inclusion-minimal elements. The exact dominance theorem ensures that every removed state has a retained replacement in every admissible context.

Corollary 9 (Exact Preservation of Existential Contextual Feasibility). *Let H be a two-terminal series--parallel graph, and let $\mathcal{C}$ be a fixed-colored two-terminal series--parallel context whose substitution by H is structurally admissible. Then the following are equivalent:*

(i) There exists an acyclic 4-edge coloring c of H such that $\mathcal{C}$ accepts (H,c).

(ii) There exists an acyclic 4-edge coloring c' of H such that $\rho(H,c') \in \mathcal{R}_4^{\min}(H)$ and $\mathcal{C}$ accepts (H,c').

Thus replacing each feasible family by its undominated family preserves existential colorability; one witness per retained state suffices for reconstruction.

Proof. Given any accepted feasible signature, choose a minimal realizable signature with the same terminal color sets and an exposed set contained in the original. Contextual dominance guarantees that the same context accepts this minimal witness. Apply the argument inductively through the decomposition tree. □

Scope: contextual necessity is proved for common admissible simple series–parallel contexts, not solely for outerplanar contexts. The reduction preserves existential feasibility and a reconstructible witness, not coloring counts; its antichain is minimal under inclusion, not claimed globally minimal over all encodings.

Figure 4 illustrates strict contextual dominance using two eight-edge cycles and a common external path C = (4, 2, 4). The same context rejects F = {{1, 3}, {2, 4}} but accepts the smaller exposed set F = {{1, 3}}.

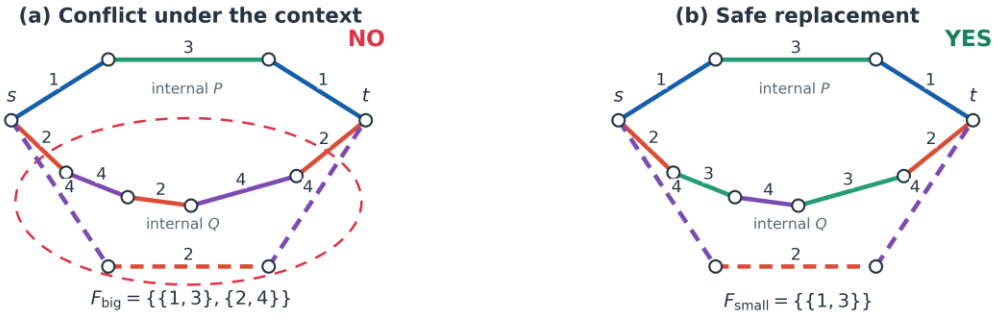


Figure 4. Strict dominance: for A(s) = A(t) = {1, 2}, the dashed context (4, 2, 4) creates a bichromatic cycle with F = {{1, 3}, {2, 4}} in (a), but not with F = {{1, 3}} in (b). Edge labels are colors.

3.4. State Counts and Algorithmic Consequences

Let $m(A,B) = |M(A,B)|$. Partition K into the disjoint subsets $X = A \cap B$, $Y = K\backslash(A \cup B)$, $U = A\backslash B$ and $V = B\backslash A$. An eligible pair consists either of one color in X and one in Y, or of one color in U and one in V. Therefore,

$$m(A,B) = |X||Y| + |U||V|. \quad (7)$$

Theorem 10 (Formal-State and Antichain Bounds). *With four colors: (i) there are at most* 16,384 *formal original signatures; (ii) there are exactly* 956 *formal exposed signatures; and (iii) every undominated feasible state family contains at most* 468 *states.*

Proof. Each terminal has 16 possible color subsets, and six unrestricted pair indicators give 16,384 formal full signatures. For fixed terminal sets, the number of exposed signatures is the number of subsets of eligible pairs. The distribution in Table 1 gives 956 formal exposed signatures.

Table 1. Distribution of eligible-pair counts for ordered terminal color sets.

$m(A,B)$	Number of (A,B)
0	60
1	48
2	120
3	16
4	12

Weighting the terminal-set-pair frequencies in Table 1 by the number of eligible subsets gives 956 formal exposed states. For each fixed terminal-set pair, Sperner's theorem [24] bounds the size of an inclusion antichain; summing these bounds yields 468. These are formal counts and worst-case bounds, not observed state counts. □

Proposition 11 (Color-Permutation Orbits). *The* 956 *formal exposed signatures form exactly* 82 *orbits under simultaneous permutations of the four colors.*

Proof. The five conjugacy classes of the four-color permutation group have respective multiplicities 1, 6, 3, 8 and 6 and fix respectively 956, 124, 28, 20 and 4 signatures. Burnside's lemma [25] therefore gives $(956 + 6\times124 + 3\times28 + 8\times20 + 6\times4)/24 = 82$ orbits. □

Color-permutation normalization is sound only if relative permutations across composed components are retained; at most 24 relative permutations arise. We therefore keep actual color labels in the implementation. The fixed finite state universe supports constant work per decomposition node.

The algorithmic novelty lies in exposed connectivity and safe contextual dominance, not in linear time alone. Formal-state bounds must not be conflated with measured runtime savings.

4. Structural Obstructions and Acyclic 4-Edge-Colorability

The results in this section concern the stated subclasses of outerplanar graphs. The graph (Q) defined below is not new: it is isomorphic to the seven-vertex graph (Q) used by Hou et al. [7] and in the edge-replacement construction discussed in the 2013 erratum of Hou and Wu [8]. An explicit isomorphism from our labeling to theirs is $0 \mapsto v,\ 1 \mapsto u,\ 2 \mapsto v',\ 3 \mapsto u',\ 4 \mapsto w',\ 5 \mapsto w,\ 6 \mapsto x$.

Accordingly, we make no novelty claim for the graph itself. Our contribution is to establish its edge-minimal failure of acyclic 4-edge-colorability through complete coloring and edge-deletion certificates (Theorem 12 and Tables 2–3), and to use this graph in the seven-vertex structural classification and the sharp threshold results for maximal outerplanar graphs (Lemma 13 and Theorem 14). These results do not establish a forbidden-subgraph characterization for all outerplanar graphs of maximum degree four.

4.1. A Seven-Vertex Obstruction and Its Certificates

A graph G is *edge-minimal non-acyclically-4-edge-colorable* if G has no acyclic 4-edge coloring but $G - e$ has one for every $e \in E(G)$. Define Q on $V(Q) = \{0,1,2,3,4,5,6\}$ with

$$E(Q) = \{01,02,05,06,15,23,24,26,34,46,56\}, \quad (8)$$

where ij denotes the edge $\{i,j\}$. The graph has eleven edges and degree sequence $(4,4,4,3,3,2,2)$. Its outer boundary is $0-1-5-6-4-3-2-0$; the remaining edges are noncrossing diagonals. Thus Q is maximal outerplanar and $\Delta(Q)=4$.

Theorem 12 (A Seven-Vertex Obstruction). *The graph Q has $a'(Q)=5$. Moreover, deleting any edge of Q produces an acyclically 4-edge-colorable graph.*

Proof of Theorem 12. The degree-four vertices $0,2,6$ form a triangle. In any proper four-edge coloring we may permute colors to assume

$$c(06)=1,\quad c(26)=2,\quad c(02)=3. \tag{9}$$

Properness at these vertices forces

$$\{c(01),c(05)\}=\{2,4\},\quad \{c(23),c(24)\}=\{1,4\},\quad \{c(46),c(56)\}=\{3,4\}. \tag{10}$$

The endpoint constraints on edges 15 and 34 leave precisely eight proper colorings. Table 2 lists their distinguishing colors and a bichromatic cycle in each. All other colors follow from (9)–(10).

Table 2. All proper four-edge colorings of Q up to permutation, and a bichromatic cycle in each.

Case	$c(01)$	$c(23)$	$c(56)$	$c(15)$	$c(34)$	Bichromatic cycle
1	2	4	3	1	2	3–2–6–4–3
2	2	4	3	1	3	4–3–2–0–5–6–4
3	4	1	4	1	2	1–0–6–5–1
4	4	4	4	1	2	1–0–6–5–1
5	4	4	3	1	2	2–0–5–6–2
6	4	4	3	1	3	2–0–5–6–2
7	4	1	4	3	2	4–2–0–1–5–6–4
8	4	4	4	3	2	2–0–6–4–2

Thus no proper four-edge coloring of Q is acyclic. For a five-color witness, take the edge ordering $(01,02,05,06,15,23,24,26,34,46,56)$ and assign, respectively, $(4,1,3,2,1,4,2,3,1,4,5)$. This assignment is proper and the subgraphs induced by every two colors are forests. Therefore $a'(Q)=5$.

Figure 5 visualizes one of the eight failing proper four-colorings in Table 2 and the valid Q − 01 certificate in Table 3. The complete impossibility proof requires all eight normalized cases.

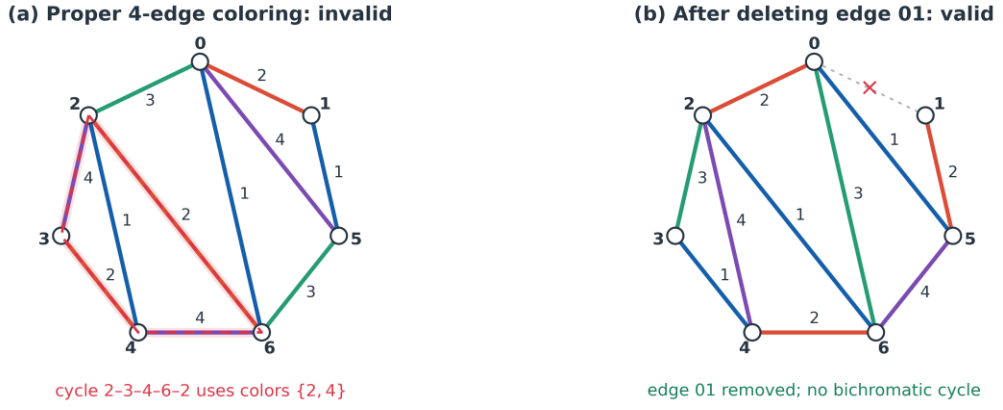


Figure 5. Seven-vertex obstruction Q: (a) a proper four-coloring containing the highlighted {2, 4}-cycle; (b) a valid acyclic four-coloring after removing edge 01 (gray dashed). Numbers denote edge colors.

To prove edge-minimality, note that Q has an automorphism interchanging $0\leftrightarrow 2$, $1\leftrightarrow 3$, and $5\leftrightarrow 4$, while fixing 6. Representatives of the six edge orbits are $01,02,05,06,15,46$. Table 3 gives acyclic four-

color witnesses for deletion of each representative, using the same edge ordering and a dash for the deleted edge.

Table 3. Acyclic four-edge coloring certificates for representative edge deletions.

Deleted edge	Colors on $(01,02,05,06,15,23,24,26,34,46,56)$
01	–, 2, 1, 3, 2, 3, 4, 1, 1, 2, 4
02	3, –, 2, 1, 1, 3, 1, 2, 4, 3, 4
05	1, 2, –, 3, 3, 4, 3, 1, 2, 4, 2
06	2, 1, 3, –, 4, 4, 2, 3, 3, 1, 2
15	4, 1, 3, 2, –, 2, 4, 3, 3, 1, 4
46	4, 1, 2, 3, 1, 3, 4, 2, 1, –, 4

The graph automorphism covers the remaining edge deletions, while Table 3 certifies the six orbit representatives. Every deletion therefore admits an acyclic four-edge coloring, proving edge-minimality. □

4.2. A Sharp Threshold in Maximal Outerplanar Graphs

A maximal outerplanar graph on at least three vertices is a triangulation of a polygon. A vertex of degree two is an *ear*, and removing an ear preserves maximal outerplanarity.

Lemma 13 (The Seven-Vertex Structure). *Every maximal outerplanar graph on seven vertices with maximum degree at most four is isomorphic to* Q.

Proof of Lemma 13. Such a graph H has $2 \cdot 7 - 3 = 11$ edges. If n_i counts the vertices of degree i, the degree sum and $n_2 + n_3 + n_4 = 7$ yield

$$2n_2 + n_3 = 6. \tag{11}$$

A maximal outerplanar graph on at least four vertices has at least two ears, so (n_2, n_3) is either $(2,2)$ or $(3,0)$.

If $n_2 = 3$, removing the three nonadjacent ears leaves a triangulated quadrilateral. Reattaching them requires three distinct boundary edges. The quadrilateral has two opposite degree-two vertices and two opposite degree-three vertices. To retain maximum degree four, the former must each receive two extra incidences, and the latter one each. But any three edges of a quadrilateral form a path whose two vertices with increment two are adjacent; the two degree-two vertices are opposite. This is impossible.

Hence $n_2 = 2$ and $n_3 = 2$. Removing both ears leaves a maximal outerplanar graph on five vertices, necessarily a fan triangulation with a central degree-four vertex. Neither ear may be reattached along an edge incident with that vertex, since the maximum degree remains four. The remaining three boundary edges form a three-edge path. Reattaching ears along two adjacent edges would increase their common degree-three vertex to five; therefore the two ears must be attached to the two nonadjacent end edges. This construction is unique up to isomorphism and yields the graph in (8). □

Theorem 14 (Sharp Maximal-Outerplanar Threshold). *Let* G *be maximal outerplanar, with* $\Delta(G) = 4$ *and* $n = |V(G)|$. *Then*

$$a'(G) = \begin{cases} 4, & n \leq 6, \\ 5, & n \geq 7. \end{cases} \tag{12}$$

Proof of Theorem 14. For $n \leq 6$, the five-vertex case has one isomorphism type, and there are two six-vertex maximal outerplanar graphs of maximum degree at most four. Color the outer edges $01,12,23,34,45,50$ and the listed diagonals as in Table 4. Each displayed coloring is proper and has no bichromatic cycle; smaller instances are obtained by restriction. Since $\Delta(G) = 4$, four colors are necessary.

Table 4. Explicit acyclic four-edge colorings of the two six-vertex maximal outerplanar graphs.

Type	Diagonals	Outer-edge colors $(01,12,23,34,45,50)$	Diagonal colors
I	15,24,25	1,2,3,1,2,4	3,4,1
II	13,15,35	3,4,2,4,1,4	1,2,3

For $n \geq 7$, repeatedly delete ear vertices until seven vertices remain. The resulting graph is maximal outerplanar, with maximum degree at most four, so Lemma 13 identifies it as Q. As Q is a subgraph of G, restriction of any acyclic coloring gives $a'(G) \geq a'(Q) = 5$. The established upper bound $a'(G) \leq \Delta(G) + 1 = 5$ for outerplanar graphs supplies the reverse inequality. □

This sharp threshold is confined to maximal outerplanar graphs; it must not be extended to the unrestricted degree-four class.

4.3. Block-Maximal Outerplanar Graphs

Definition 15 (Block-Maximal Outerplanar Graph). *An outerplanar graph is block-maximal outerplanar if each block with at least three vertices is maximal outerplanar on its vertex set. Single-edge blocks are allowed.*

Theorem 16 (Exact Blockwise Criterion). *Let G be block-maximal outerplanar with $\Delta(G) = 4$. Then $a'(G) = 5$ if and only if some block of G contains at least seven vertices. Equivalently, $a'(G) = 4$ if and only if every block has at most six vertices.*

Proof. Suppose first that some block B has at least seven vertices. By the definition of block-maximal outerplanar graphs, B is maximal outerplanar. Hence $|E(B)| = 2|V(B)| - 3$, and its average degree is

$$\bar{d}(B) = \frac{2|E(B)|}{|V(B)|} = 4 - \frac{6}{|V(B)|} > 3$$

Since $\Delta(B) \leq \Delta(G) = 4$, it follows that $\Delta(B) = 4$. Theorem 14 therefore gives $a'(B) = 5$. Since B is a subgraph of G, we have $a'(G) \geq a'(B) = 5$. The outerplanar upper bound $a'(G) \leq \Delta(G) + 1 = 5$ then yields $a'(G) = 5$.

Conversely, suppose that every block has at most six vertices. A bridge block is trivially acyclically 4-edge-colorable. For any block (B) with at least three vertices, B is maximal outerplanar. If $\Delta(B) = 4$, Theorem 14 gives $a'(B) = 4$. If $\Delta(B) \leq 3$, the standard outerplanar bound gives $a'(B) \leq \Delta(B) + 1 \leq 4$). Thus, every block admits an acyclic 4-edge coloring. By Lemma 1, these colorings can be combined to obtain an acyclic 4-edge coloring of G. Since $\Delta(G) = 4$, at least four colors are necessary, and hence $a'(G) = 4$. □

5. Exact Algorithm and Computational Validation

5.1. Recognition, Reconstruction and Complexity

Represent each terminal color set by four bits and each exposed connectivity set by six bits. For each block, propagate realizable signatures bottom-up through a binary two-terminal series–parallel decomposition, applying Eqs. (2)–(4) and discarding duplicates and dominated signatures. Store one compatible child-state pair for every retained parent state.

5.1.1. State Propagation and Witness Reconstruction

Each edge leaf has four possible color assignments. At an internal node, combine all retained child states by the exact series/parallel rules and retain only inclusion-minimal exposed sets for each ordered terminal-set pair. If the root family is empty, return NO; otherwise follow witness pointers to reconstruct a

valid four-edge coloring. Across the block–cut tree, permute each new block's color labels to avoid conflicts at articulation vertices; this introduces no cycle spanning blocks.

5.1.2. Correctness and Complexity

Theorem 17 (Exact Recognition and Reconstruction). For every finite simple outerplanar graph of maximum degree at most four, the dominance-reduced dynamic program returns YES exactly when an acyclic four-edge coloring exists; in the YES case its witness reconstruction returns such a coloring.

Proof. At every leaf the algorithm enumerates all four possible colors. Exact composition generates only realizable signatures, and the contextual dominance theorem replaces any omitted state by a retained compatible one. Structural induction therefore proves nonemptiness of a block's root family exactly when the block is four-colorable. Witness pointers realize the chosen states. Finally, the block reduction combines these proper acyclic colorings by color permutation at articulation vertices. □

Proposition 18 (Linear-Time Bound). Given linear-size valid two-terminal series–parallel decompositions of all nontrivial blocks, the algorithm recognizes acyclic four-edge colorability and reconstructs a witness, when one exists, in linear time and space.

Proof. The fixed four-color universe contains at most 468 undominated formal states (Theorem 10). Consequently, each decomposition node performs bounded pairwise transitions and dominance checks, storing a bounded number of witness pointers. There are linearly many nodes and the block–cut forest is computable in linear time [26]; reconstruction visits each selected pointer once. Thus the total time and space are linear in graph size. □

5.2. Benchmark Design

We evaluated 649 benchmark entries: 332 outerplanar Graph Atlas graphs with at most seven vertices, maximum degree at most four, and at least one edge, five representative maximal outerplanar graphs, 200 nonmaximal and 100 multiblock generated graphs, and 12 targeted obstruction/deletion cases. The random groups comprise 40 nonmaximal and 20 multiblock instances at each order $n \in \{12,24,48,96,192\}$, yielding 60 generated instances per order. Isomorphic duplicates were excluded within each group and order. The random generator used seed 20261106. All tests satisfy the stated degree and outerplanarity constraints; targeted cases can overlap with the atlas set.

Python 3.13.5 and NetworkX 3.6.1 [27] were used on a Core i9-14900HX laptop (32 GB RAM). Full-DP, Exposed-DP and Dominance-DP used identical instances, decompositions and color labels. For each instance and DP variant, one untimed warm-up run was performed, followed by seven timed runs. The per-instance DP execution time was recorded as the median of the seven measurements, using perf_counter_ns() to measure the run_tree operations summed over all blocks. Graph decomposition, validation, and memory tracing were excluded from the timing measurements. Peak traced Python memory was measured in a separate execution using Python's tracemalloc module and reported in decimal megabytes (MB); it does not represent whole-process resident memory (RSS).

5.3. Exactness and Cross-Algorithm Validation

The three DP variants were evaluated using identical decomposition trees and color labels. At every internal decomposition node (x), the computed signature families were compared to verify the following identities:

$$S_x^E = \pi(S_x^F), \qquad S_x^D = \mathrm{Min}(S_x^E)$$

At every decomposition node, the Exposed-DP states were checked against projection of Full-DP, and Dominance-DP against the inclusion-minimal projected family. Independent exhaustive backtracking gave 331 YES and one NO among 332 atlas graphs; all 12 targeted obstruction cases and reconstructed colorings were checked for properness and absence of bichromatic cycles. Table 5 summarizes the consistency tests.

Table 5. Correctness and cross-algorithm consistency results.

Benchmark	Instances	Reference YES/NO	DP disagreements	State / certificate errors
Graph Atlas	332	331 / 1	0	0 / 0
Obstruction tests	12	11 / 1	0	0 / 0
Other benchmarks	305	—	0	0 / 0
Total	649	—	0	0 / 0

Note: Counts refer to benchmark entries, not necessarily pairwise nonisomorphic graphs; targeted obstruction cases may overlap with Graph Atlas. A dash indicates no independent exhaustive reference count.

5.4. State Reduction and Computational Trade-offs

We evaluated three exact dynamic programming variants—Full-DP, Exposed-DP, and Dominance-DP—using identical instances and decomposition trees. Exposed-DP removes terminal-blocked connectivity information, while Dominance-DP further eliminates inclusion-dominated states according to Theorem 7.

$$S_v(G) = \sum_{x \in T_G} |\mathcal{F}_v(x)|.$$

where T_G is the decomposition tree and $\mathcal{F}_v(x)$ denotes the retained state family at node x.

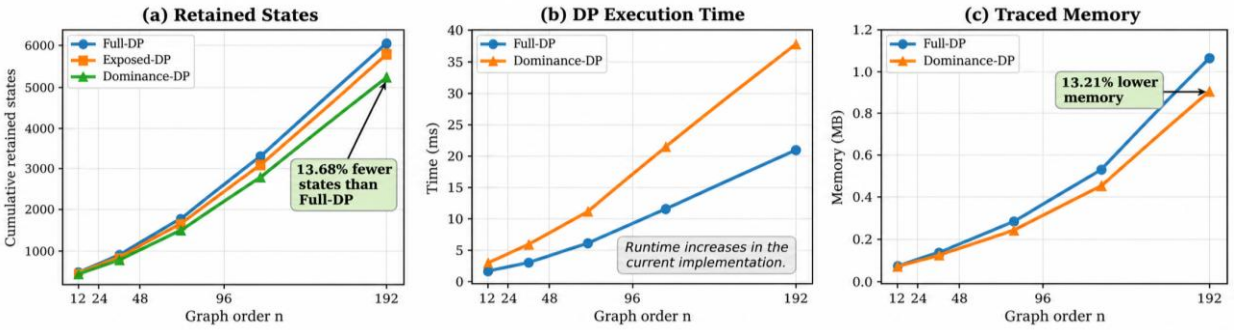


Figure 6. State reduction and computational trade-offs across graph orders $n \in \{12,24,48,96,192\}$: (a) mean cumulative retained states; (b) mean DP-only execution time; and (c) mean peak traced Python memory. Each point reports the unweighted mean over 60 generated instances (40 nonmaximal and 20 multiblock); execution times are per-instance medians of seven timed runs, and memory values are separately measured peak tracemalloc allocations.

Figure 6 shows the computational trade-offs. At (n=192), Dominance-DP reduced mean retained states by 13.68% and peak traced memory by 13.21% relative to Full-DP. However, execution time increased from 20.68 ms to 38.09 ms, indicating additional dominance-maintenance overhead in the current implementation.

5.5. Application to Pairwise-Acyclic Link Scheduling

Acyclic edge coloring provides a natural graph-theoretic model for constrained link scheduling in sparse communication networks. Consider a network $G = (V, E)$, where vertices represent communication devices and edges represent links. A k-slot assignment is a mapping $c: E \to \{1,2,\ldots,k\}$. Let $E_i = \{e \in E: c(e) = i\}$. A feasible schedule requires two conditions: $c(e) \neq c(f)$, whenever $e \neq f$ and $e \cap f \neq \emptyset$, $G[E_\mathrm{i} \cup E_\mathrm{j}]$ is a forest, for all $1 \leq i < j \leq k$.

The first condition prevents incident links from sharing a slot, whereas the second excludes bichromatic cycles between any pair of slots. Thus, a feasible schedule is precisely an acyclic edge coloring.

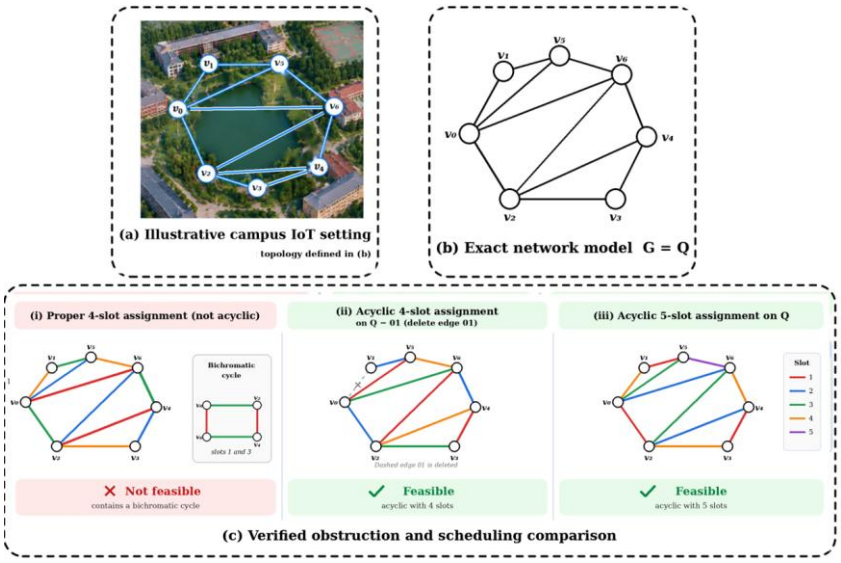


Figure 7. Application of acyclic edge coloring to pairwise-acyclic link scheduling. (a) Conceptual campus IoT network setting. (b) Exact seven-vertex outerplanar graph Q. (c) Scheduling comparison: (i) a proper but non-acyclic four-slot assignment containing a bichromatic cycle; (ii) a feasible acyclic four-slot assignment on Q − v_0v_1; and (iii) a feasible acyclic five-slot assignment on Q. The dashed edge indicates the deleted link.

Figure 7 illustrates this model using a seven-vertex outerplanar network Q with maximum degree four. The campus IoT scene in Fig. 7(a), while Fig. 7(b) specifies the exact network topology. As shown in Fig. 7(c-i), a proper four-slot assignment may still be infeasible because slots 1 and 3 form the bichromatic cycle v_0–v_2–v_4–v_6–v_0. By Theorem 16, Q admits no acyclic four-slot assignment. Removing edge v_0v_1 yields a feasible acyclic four-slot assignment on Q − v_0v_1, as shown in Fig. 7(c-ii), whereas Fig. 7(c-iii) provides a feasible acyclic five-slot assignment on the original graph. This experiment shows that the proposed exact algorithm can distinguish ordinary slot feasibility from pairwise-acyclic scheduling and reconstruct valid assignments satisfying the given graph constraints.

5.6. Deployment-Inspired Multi-Floor Link Scheduling

To further illustrate practical applicability, we constructed a synthetic two-floor IoT network comprising 71 vertices and 76 links (Fig. 8). The network is connected and outerplanar, with maximum degree four and six independent cycles. The four-slot assignment was verified to contain neither adjacent-link conflicts nor bichromatic cycles across all six color pairs. Hence, the instance satisfies $a'(G) = 4$, demonstrating feasible pairwise-acyclic scheduling in a larger cyclic network. The topology is deployment-inspired rather than derived from field measurements.

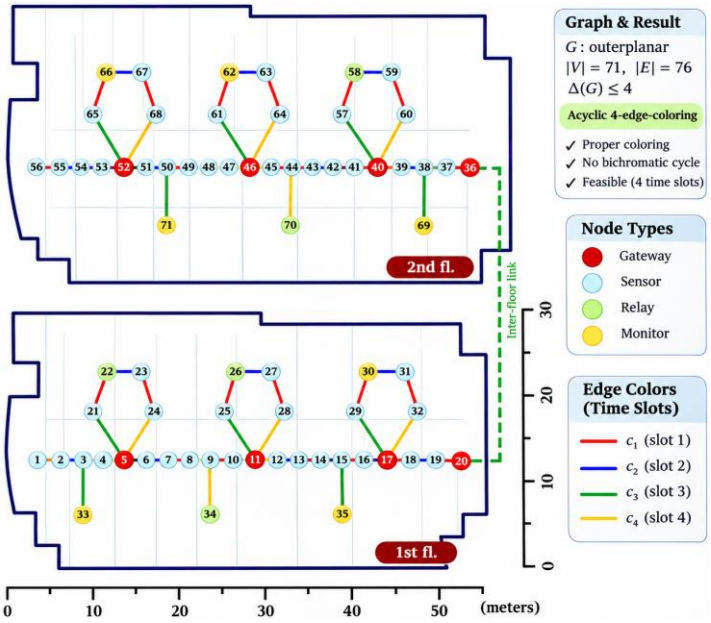


Figure 8. Deployment-inspired two-floor IoT communication network with a verified acyclic four-slot link schedule. Numbered vertices represent communication devices, and red, blue, green, and yellow edges indicate slots 1–4, respectively. The dashed green link connects the two floors. The 71-vertex, 76-edge outerplanar network satisfies $\Delta(G) = 4$, with no adjacent-link conflicts or bichromatic cycles.

6. Conclusions

This work develops an exact exposed-signature framework for acyclic four-edge coloring of outerplanar graphs with maximum degree at most four. Exact composition laws and contextual dominance establish that inclusion-minimal feasible states preserve both existence and reconstruction. The formal state space decreases from 16,384 to 956 signatures, with at most 468 undominated states, supporting linear-time recognition given suitable series–parallel decompositions. Verified certificates identify an edge-minimal seven-vertex obstruction and sharp criteria for maximal and block-maximal subclasses. Experiments across 649 entries confirm implementation consistency and demonstrate the computational trade-offs. Further investigation should address broader graph classes.

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Data Availability Statement: The source code, benchmark graph instances, experimental measurements, computational verification certificates, and reproducibility scripts supporting the findings of this study are publicly available in the GitHub repository at https://github.com/caozei8543/outerplanar-acyclic-4-edge-coloring.

Ethics Approval: Not applicable.

Conflicts of Interest: The author declares no conflicts of interest.

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